Regression and Latent Variable Models for Bounded Outcomes

Jorge Luis Bazán*
University of São Paulo

Abstract

Bounded outcomes arise naturally in many areas of scientific, social, biomedical, psychological, and educational research. Proportions, percentages, rates, indices, Likert-type responses, test scores, item-level responses, and diagnostic classifications are common examples of variables whose possible values are restricted to a continuous finite interval, such as $(0, 1)$, $[0, 1]$, or another bounded range including $\{0, 1, \ldots, n\}$.

This short course introduces modern regression strategies for analyzing bounded outcomes, with emphasis on flexible model formulation, interpretation, estimation, and applied implementation. The course will cover five main topics:

- bounded regression models for continuous responses restricted to finite intervals [1];
- bounded count models for discrete outcomes with known upper limits [2];
- bounded item response models for educational and psychological measurement [3]
- bounded cognitive diagnostic models for classifying latent attributes under constrained response structures [4]

Participants will learn how these models are constructed, how their parameters can be estimated and interpreted, and how these proposal model are alternatives to modeling of datasets. Special attention will be given to choosing an appropriate model according to the nature of the outcome, the presence of boundary values, the measurement scale, and the substantive research question.

Throughout the course, methodological concepts will be connected to practical examples from different fields, including education, health sciences, behavioral research, and social indicators. Participants will work with examples involving proportions, restricted scores, bounded counts, and item-based measurements, gaining experience with both conceptual interpretation and implementation considering recent authored publications.

The course is designed for researchers, graduate students, and applied analysts who need principled tools for modeling outcomes that cannot be adequately analyzed using unbounded regression methods.

In addition to applied training, the course will highlight open methodological questions and possible directions for future research. Topics such as model extensions, computational

Email: jlbazan@icmc.usp.br.

challenges, latent-variable formulations, diagnostic accuracy, and the integration of bounded regression with measurement models will be discussed. By the end of the course, participants will have a broad understanding of the main families of bounded outcome models and the practical skills needed to apply them in empirical research.

Keywords: Bounded distributions, Bounded mixed regression, Bounded Count regression, Bounded Item response model, Bouded Cognitive Diagnostic Model.

References

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[3] Molenaar, D., Cúri, M., & Bazán, J. L.,*Zero and one inflated item response theory models for bounded continuous data*, Journal of Educational and Behavioral Statistics, (2022) 47(6), 693-735. Click here

[4] de Oliveira, E. S., Wang, X., Bazán, J. L., & de La Torre, J.,*A cognitive diagnosis model for latent classification of bounded continuous variables*, British Journal of Mathematical and Statistical Psychology, (2026) 47(6), 693-735. Click here